

Hamromaster

COMPLETE NOTES

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**For Hamromaster Online
Classes**

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Probability: distribution ← 1 Long Question

arrangement → variable (सी)

→ सामान्यतः prob. distribution frequency distribution है।



→ A probability distribution is the mathematical function that gives the probabilities of occurrence of different possible outcomes for an experiment. It is a mathematical description of a random phenomenon in terms of its sample space and the probabilities of events.

→ A prob. distribution is a statistical function that describes all the possible values and likelihood that a random variable can take within a given range.

→ As frequency distribution of the variable, we can distribute total prob. to different mass points in case of discrete random variables or to different class interval in case of continuous random variables.

→ A prob. distribution possesses all the characteristics of an observed frequency distribution. we can define popⁿ mean, popⁿ standard deviation as in frequency distribution.

→ A prob. distribution for a particular random variables is a function or table of values that...

If the variable takes all possible value within a certain interval then the random variable is called a Continuous random Variable. The amount of rainfall on a rainy Season is example of continuous random Variables.



Binomial Distribution

Binomial distribution is a discrete prob. distribution named after the name of its propounder "James Bernoulli" (1654-1705). He propounded Bernoullian trials which is a process where an experiment is performed repeatedly resulting either a 'Success' or a 'failure' in each trial. Binomial distribution enlarged form of Bernoulli distribution under following assumptions.

- i) Each trial has only two possible outcomes called 'Success' or 'failure'.
- ii) prob. of success remains same from trial to trial.
- iii) Trials are independent and experiment is performed randomly for finite number.

प्रश्न-चिह्न आकार

- जहाँले पनि दिइएको prob. of success (p) 5% वटि र n , 20 भन्दा कम हुनुपर्दछ।
- only two outcomes.
- Number 'n' of trial is finite.
- All the trials are independent.

defⁿ:

A discrete random variable 'x' which has the following probability mass function is

$$P(X=x) = {}^n C_x \cdot p^x \cdot q^{n-x}, \quad x = 0, 1, \dots, n. \text{ is}$$

Binomial variate and its distribution is called Binomial distribution. It is denoted by $X \sim B(n, p)$

यहाँ n & p binomial distribution के parameter हैं। परसेगरी

x = Success event

p = prob. of success

q = prob. of failure.

$${}^n C_r = \frac{n!}{r!(n-r)!}$$

$$0! = 1$$

$$n! = n(n-1)!$$

...

$$p+q=1$$



characteristics of the binomial dist...

- Since it assumes only the positive integers value like $0, 1, 2, \dots, n$, binomial distribution is a discrete prob. distribution.
- Binomial distribution is known as bi-parametric distribution as it is characterized by two parameters n and p .

The mean of binomial distribution is given by

$$\bar{x} = E(X) = np$$

$$\text{Variance } \text{Var}(X) = \sigma_x^2 = npq$$

$$\text{Skewness} = \frac{(q-p)^2}{npq}, \quad \text{kurtosis} = 3 + \frac{1-6pq}{npq}$$

✓

कहें/प्रश्न हकत हैरी

- Four coins are tossed simultaneously. what is the prob. of getting: (a) exactly 3 heads, (b) at most two heads, (c) at least one head (d) less than two heads (e) not less than three heads (f) more than one head (g) between 1 & 4 heads?



Solⁿ

Here,

AS per the information of coin toss given in the question, it belongs to the binomial distribution.

we have,

$$n = \text{No. of coins tossed} = 4$$

$$p = \text{prob. of head} = 0.50$$

$$q = \text{prob. of tail} = 1 - 0.50 = 0.50$$

Let x be the no. of heads, then the prob. of getting ' x ' heads under binomial of getting ' x ' heads under binomial distribution is given by

written by
palpali bharat

$$P(x) = {}^n C_x \cdot p^x \cdot q^{n-x}$$

Now,

(a) Exactly 3 heads ($x=3$)

$$P(3) = {}^4 C_3 (0.5)^3 (0.5)^{4-3} = 0.25$$

(b) At most 2 heads, (वर्किसा 2)

$$P(r \leq 2) = P(X=0) + P(X=1) + P(X=2) \quad | \quad r = \overleftarrow{0, 1, 2, 3, 4}$$

OR

$$= 1 - [P(X=3) + P(X=4)]$$

$$= 1 - (0.25 + 0.0625)$$

$$= 0.6875$$



(c) At least 1 heads (~~कमिमा~~ 1 head)

$$(r \geq 1)$$

$$r = 0, \overbrace{1, 2, 3, 4}^{P=?}$$

we have,

$$P(0) = {}^4C_0 (0.50)^0 (0.50)^4 = 0.0625$$

Now,

$$\begin{aligned} \text{Req. prob.} &= 1 - P_0 \\ &= 1 - 0.0625 \\ &= 0.9375 \end{aligned}$$

(d) Less than 2 heads ($r < 2$)

$$r = 0, \overleftarrow{1}, \overline{2, 3, 4} \quad \leftarrow P=?$$

we have,

$$P(0) = 0.0625$$

$$P(1) = {}^4C_1 (0.50)^1 (0.50)^3 = 0.25$$

Now,

$$\begin{aligned} \text{Req. prob.} &= P_0 + P_1 \\ &= 0.0625 + 0.25 \\ &= 0.3125 \end{aligned}$$

(e) Not less than 3 heads (3 भन्दा कम होइन)

$$(r \geq 3)$$

$$r = 0, 1, 2, \overset{p=?}{\rightarrow} 3, 4$$

we have,

$$P(3) = 0.25$$

$$P(4) = 0.0625$$



$$\begin{aligned}\therefore P(r \geq 3) &= P(3) + P(4) \\ &= 0.25 + 0.0625 \\ &= 0.3125\end{aligned}$$

(f) More than 1 heads ($r > 1$)

$$r = 0, 1, \overset{p=?}{\rightarrow} 2, 3, 4$$

we have,

$$P(0) = 0.0625$$

$$P(1) = 0.25$$

Now,

$$\begin{aligned}\text{Req. prob.} &= 1 - (P_0 + P_1) \\ &= 1 - (0.0625 + 0.25) \\ &= 1 - 0.3125 \\ &= 0.6875\end{aligned}$$

(g) Between 1 & 4 heads ($1 < r < 4$)

$$r = 0, 1, \overset{p=?}{\rightarrow} 2, 3, 4$$

we have,

$$P(2) = {}^4C_2 (0.5)^2 (0.5)^2 = 0.375$$

$$P(3) = 0.25$$

$$\therefore P(1 < r < 4) = P(2) + P(3) \Rightarrow 0.625$$

(Q2) A Survey conducted in 2018 found about 30% of the Nepalese workers who have gone abroad are illegal. A sample of 7 Nepalese workers are randomly picked, find the prob. of at least two are legal.

Solⁿ

Here,

Here $p > 5\%$ & $n < 20$. So it belongs to the binomial distribution.

we have,

$$p = \text{prob. of legal workers} = 1 - 0.30 \Rightarrow 0.70$$

$$q = \text{prob. of illegal workers} = 0.30$$

$$n = \text{No. of selected workers} = 7$$

Let 'r' be the no. of legal workers, then the prob. of getting 'r' under binomial distribution is given by:

$$P(r) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

At least 2 legal ($r \geq 2$)

$$r = 0, 1, \overbrace{2, 3, \dots, 7}^{\rightarrow}$$

we have,

$$P(0) = {}^7C_0 (0.7)^0 (0.3)^7 \Rightarrow 0.0002$$

$$P(1) = {}^7C_1 (0.7)^1 (0.3)^6 \Rightarrow 0.0036$$

$$\begin{aligned} \text{Now, Req. prob.} &\Rightarrow 1 - (P_0 + P_1) \\ &= 1 - (0.0002 + 0.0036) \\ &= 0.9962 \quad \checkmark \end{aligned}$$



1) There are 200 staffs in a municipality office of which 120 staffs are female. If 10 staffs are selected for a training session, find the prob. of selecting at least one male.

यदि यदि प्रश्न के अनुसार data दी गई है और तबसात prob. find करने के लिये prob. हैरी Binomial distribution है कि दोन और सुनिश्चित हुनु पर्दे। साथै जुन data वाट prob. निश्चित्यकी हुन्छ तबसात कदापी 'n' मान्न सकिन्छ

Solⁿ

Here,

200 $\rightarrow$ 120 female
 $\rightarrow$ 80 male.



As per the information given in the question we need to apply binomial distribution for required probability.

we have,

$$p = \text{prob. of male Staffs} = \frac{80}{200} = 0.40$$

$$q = \text{prob. of female Staffs} = 1 - 0.40 = 0.60$$

$$n = \text{No. of selected staffs} = 10.$$

$$P(r) = {}^n C_r \cdot p^r \cdot q^{n-r}$$

Now,

At least 1 male ($r \geq 1$) $P = ?$

$$r = 0, 1, 2, \dots, 10$$

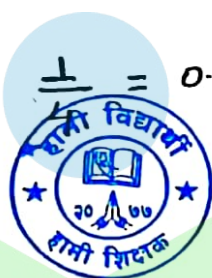
$$\begin{aligned} \therefore \text{Req. Prob.} &= 1 - P(0) \\ &= 1 - {}^{10} C_0 \cdot (0.4)^0 \cdot (0.6)^{10} \\ &= 1 - 0.006 \\ &= 0.994 \end{aligned}$$

(4) A multiple choice test has 5 questions and there are four choices for each question. The test is conducted in a group of 200 students. Find the number of students who answers:

(a) Three question correct

(b) No. question correct

1.	a	b	c	d
2.	a	b	c	d
3.	a	b	c	d
4.	a	b	c	d
5.	a	b	c	d



$\frac{1}{4} = 0.25 \Rightarrow$ prob. of correct answer. ✓

Soln:

$$n = \text{No. of question} = 5$$

$$p = \text{prob. of correct} = \frac{1}{4} = 0.25$$

$$q = 1 - p \Rightarrow 1 - 0.25 \Rightarrow 0.75$$

$$N = \text{Total Students} = 200$$

$$\therefore P(X) = {}^n C_r \cdot p^r \cdot q^{n-r}$$

$$\begin{aligned} \text{(a)} \quad P(r=3) &= {}^5 C_3 (0.25)^3 (0.75)^{5-3} \\ &= 0.0878 \end{aligned}$$

$\therefore$ The number of students who answer three question correct = $N \times P(r)$

$$\begin{aligned} &= 200 \times 0.0878 \\ &= 17.58 \approx 18 \text{ students} \end{aligned}$$

(b) No. question correct ($r=0$)

∴ The number of students who answers no question correct = $N \times p(x)$

$$= 200 \times 0.2372$$

$$= 47.46$$

$$\approx 47 \text{ students} \checkmark$$

(5) Find the prob. of getting 2 success, if the mean and standard deviation of the binomial distribution is 5 and 2 respectively
Solⁿ

Here,

$$\text{mean} = np = 5 \quad \text{--- (i)}$$

$$SD = \sqrt{npq} = 2$$

$$\therefore npq = 4 \quad \text{--- (ii)}$$

from eqⁿ (ii),

$$5 \times q = 4$$

$$\therefore q = 4/5 \Rightarrow 0.80$$

$$\text{Then, } p = 1 - q \Rightarrow 1 - 0.80 \\ \Rightarrow 0.20$$

from eqⁿ (i),

$$n \times 0.20 = 5$$

$$n = 5/0.20 \Rightarrow 25$$

The prob. under binomial distribution can be calculated as



$$P(r) = {}^n C_r p^r q^{n-r}$$

Getting 2 success ($r=2$)

$$P(2) = {}^{25} C_2 (0.20)^2 (0.80)^{23}$$

$$= 0.071$$



fit Binomial distribution

X	0	1	2	3	4
frequency (f)	10	25	30	20	15

Solⁿ:

$$N = \sum f = 100$$

Let p be the prob. of success. Then

$$\text{mean} = np$$

$$\text{or, } \bar{x} = np$$

$$p = \frac{\bar{x}}{n} \Rightarrow \frac{\sum fx}{nN}$$

$$= \frac{0 \times 10 + 1 \times 25 + 2 \times 30 + 3 \times 20 + 4 \times 15}{4 \times 100}$$

$$= 0.5125$$

Let x denote number of successes
then expected frequency $f(x)$ can be
calculated as,

$$f(x) = N p(x)$$

$$= N {}^n C_r p^r q^{n-r}; \quad r = 0, 1, 2, 3, 4$$

$$= 100 \times {}^4 C_r p^r (1-p)^{4-r}$$

$$= 100 \times {}^4 C_r (0.5125)^r (1-0.5125)^{4-r}$$

Expected frequencies for $r = 0, 1, 2, 3, 4$ are tabulated

r	observed frequency	Expected frequency. $f(r) = N p(r)$
0	10	$100 \times {}^4 C_0 (0.5125)^0 (0.4875)^{4-0} = 5.62 \approx 6$
1	25	$100 \times {}^4 C_1 (0.5125)^1 (0.4875)^{4-1} = 23.75 \approx 24$
2	30	$100 \times {}^4 C_2 (0.5125)^2 (0.4875)^{4-2} = 37.45 \approx 37$
3	20	$100 \times {}^4 C_3 (0.5125)^3 (0.4875)^{4-3} = 26.25 \approx 26$
4	15	$100 \times {}^4 C_4 (0.5125)^4 (0.4875)^{4-4} = 6.89 \approx 7$
	$N = 100$	Total Expected frequency $= 100$

Nature of coin

$$p = q = 0.5$$

'p' should be calculated from following relationship

$$\text{Mean } (np) = \frac{\sum Fx}{N}$$

If question remains silent about the nature of coins, then in that case again we $p = q = 0.50$

Q. find value of q in binomial distribution of 6 independent trials if prob. of getting 2 successes is equal to q times of prob. of getting 4 successes.

soln

Here, Binomial distribution with n trials & x success is

$$P(x) = {}^n C_x \cdot p^x \cdot q^{n-x}, \quad x = 0, 1, 2, \dots, n$$

we are given, $n = 6$

$$\text{Also, } 9P(x=4) = P(x=2)$$

$$\text{or, } 9 \times \frac{{}^6 C_4}{4!2!} p^4 (1-p)^2 = \frac{{}^6 C_2}{4!2!} (p^2) (1-p)^4$$

$$\text{or, } 9 \times p^2 = (1-p)^2$$

$$\text{or, } 9p^2 = 1 - 2p + p^2$$

$$\text{or, } 8p^2 + 2p - 1 = 0$$

$$\text{or, } (4p-1)(2p+1) = 0$$

$$\text{or, either, } 4p-1=0 \quad \text{or } 2p+1=0$$

$$\therefore p = \frac{1}{4} \quad \text{or } p = -\frac{1}{2} \text{ (which is impossible)}$$

$$\therefore p = \frac{1}{4} \quad \text{and hence } q = 1-p$$

$$\therefore q = 1 - \frac{1}{4} \Rightarrow \frac{3}{4} \quad \checkmark$$



Normal Distribution

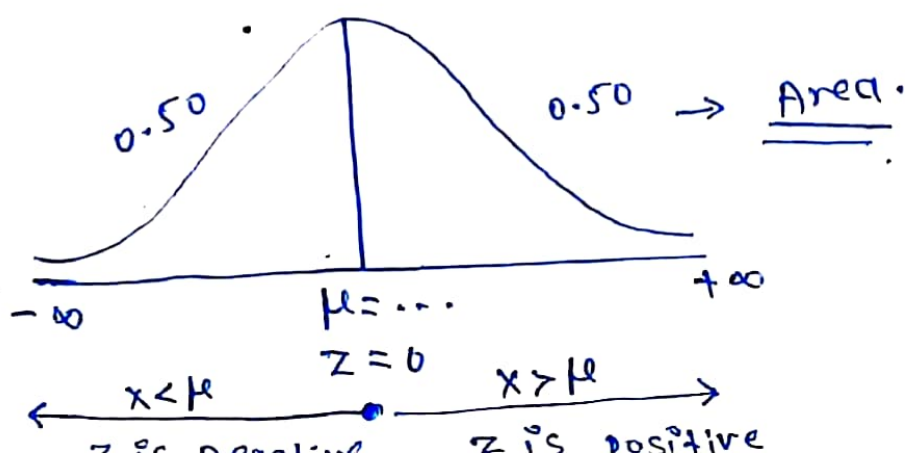
- It is the continuous probability distribution.
- It is the most popular and commonly used Probability model in Statistical analysis.
- It generally fits on common statistical data relating to the economics, business, social science and physical science as well.

It was first discovered by an English mathematician De-moivre in 1733 and later it was developed by Carl Friedrich Gauss in 1809.

In order to determine the required probability under normal distribution, we should calculate the Standard normal variate (i.e. z -value) and that z value should be plotted into the normal curve.

Formula

$$Z = \frac{X - \mu}{\sigma} \rightarrow \text{Always keep 2 digits decimal.}$$



where,

x = value of random variable

μ = mean of normal distribution.

σ = Std. deviation of normal distribution.

Note:

z table मा दिइएको prob. value जहिले पनि
 x देखि μ सम्मको हुन्छ ।

Cases in Normal distribution

	x	x			(Imp)
Parameters	case I	case II	case III	IV	V
μ	✓	✓	✓	?	?
σ	✓	✓	?	✓	?
x	✓	?	✓	✓	✓
p	?	✓	✓	✓	✓



defⁿ:

Ques-1
The income group of 10,000 persons was found to be normally distributed with a mean RS 750 and a Standard deviation RS 50. Find

- (a) the percentage of the person whose income is less than RS. 800.
- (b) the percentage of the person whose income is less than RS 710.
- (c) the percentage of the persons whose income is above RS. 688.
- (d) the number of the person whose income is between RS 714 and RS 822.
- (e) the number of the person whose income is between RS 675 and RS 725.

Solⁿ:

Given that,

Total persons (N) = 10,000

mean income (μ) = RS 750

Std. Deviation (σ) = RS 50

we know,

$$Z = \frac{X - \mu}{\sigma}$$

Req. (a) less than RS 800 ($X < 800$)

Taking $X = 800$

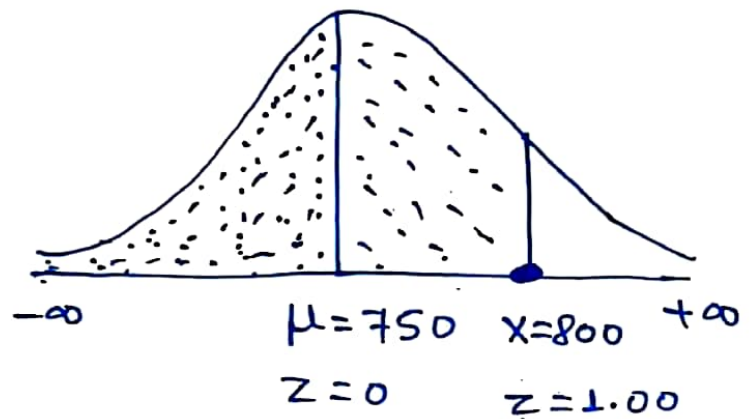


यहाँ percentage
सॉल्विंग करने की
Prob. दी।

$$Z = \frac{X - \mu}{\sigma}$$

$$= \frac{800 - 750}{50}$$

$$= 1.00$$



Now,

$$P(X < 800) = P(Z < 1.00)$$

$$= P(-\infty < Z < 1.00)$$

$$= P(-\infty < Z < 0) + P(0 < Z < 1.00)$$

$$= 0.5 + 0.3413$$

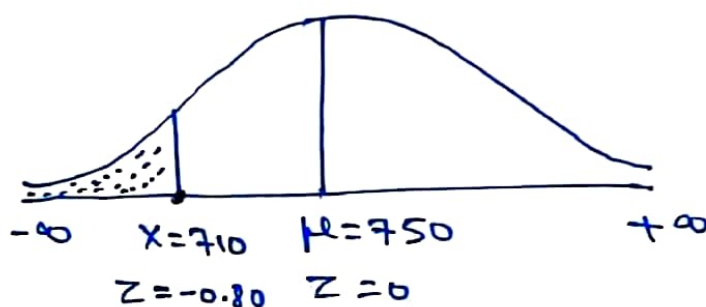
$$= 0.8413$$

(~~By symmetry~~)

⑥ less than RS 710 ($X < 710$)

Taking $X = 710$,

$$Z = \frac{X - \mu}{\sigma} = \frac{710 - 750}{50} = -0.80$$



Now,

$$\begin{aligned} P(X < 710) &= P(Z < -0.80) \\ &= P(-\infty < Z < 0) - P(-0.80 < Z < 0) \\ &= P(0 < Z < \infty) - P(0 < Z < 0.80) \\ &\quad \text{By Symmetry.} \\ &= 0.5 - 0.2881 \\ &= 0.2119 \end{aligned}$$

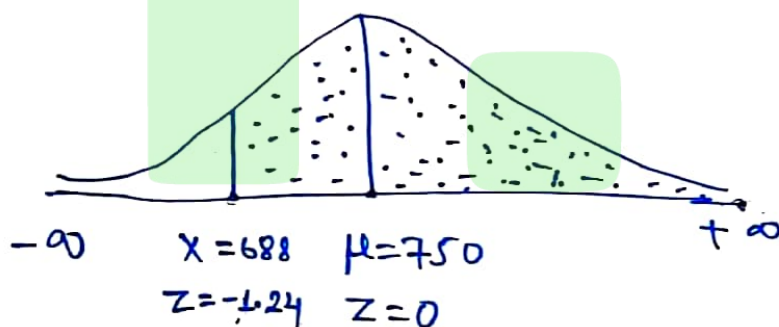


Req. (c)

Above Rs 688 ($X > 688$)

Taking $X = 688$

$$Z = \frac{X - \mu}{\sigma} = \frac{688 - 750}{50} = -1.24$$



Now,

$$P(X > 688) = P(Z > -1.24)$$

$$= P(-1.24 < Z < \infty)$$

$$= P(-1.24 < Z < 0) + P(0 < Z < \infty)$$

$$= P(0 < Z < 1.24) + P(0 < Z < \infty)$$

By Symmetry

$$= 0.3925 + 0.5$$

$$= 0.8925$$

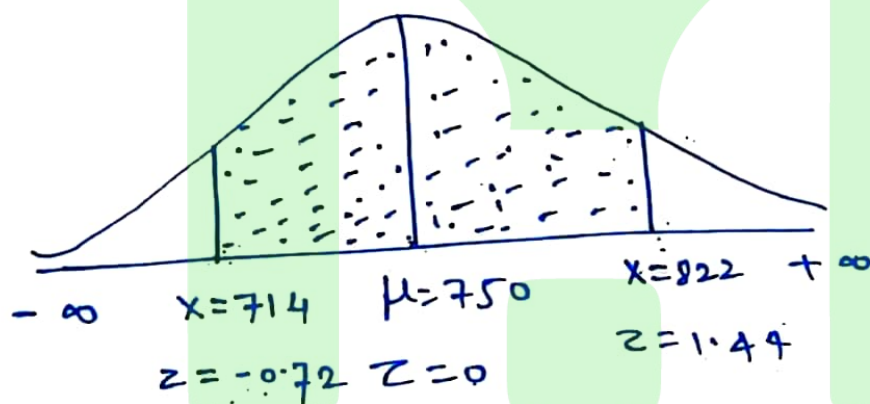
(d) Between 714 and 822 ($714 < X < 822$)

Taking $X = 714$

$$z = \frac{X - \mu}{\sigma} = \frac{714 - 750}{50} = -0.72$$

Taking $X = 822$

$$z = \frac{X - \mu}{\sigma} = \frac{822 - 750}{50} = 1.44$$



Now,

$$P(714 < X < 822)$$

$$= P(-0.72 < Z < 1.44)$$

$$= P(-0.72 < Z < 0) + P(0 < Z < 1.44)$$

$$= P(0 < Z < 0.72) + P(0 < Z < 1.44) \quad (\text{By symmetry})$$

$$= 0.2642 + 0.4251$$

$$= 0.6893$$

The number of the person whose income is between Rs 714 and Rs 822

$$= N \times p$$

$$= 10,000 \times 0.6893$$

$$= 6893 \text{ persons.}$$



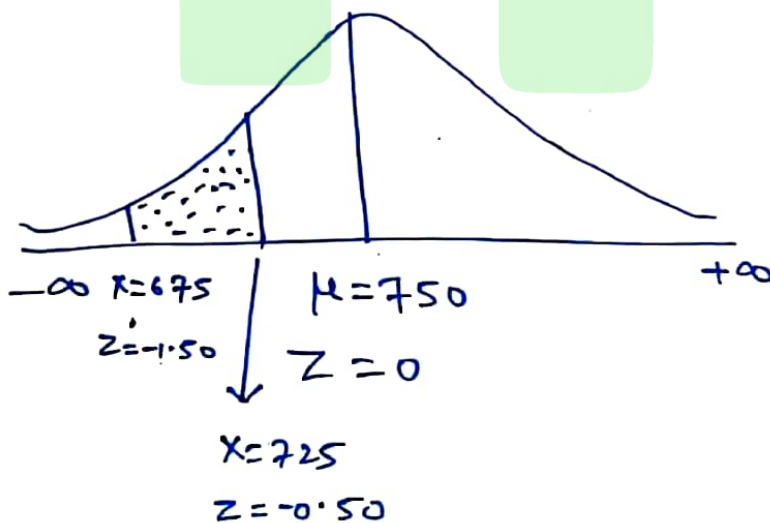
Req. e) Between 675 and 725 ($675 < X < 725$)

Taking $X = 675$

$$z = \frac{X - \mu}{\sigma} = \frac{675 - 750}{50} = -1.50$$

Taking $X = 725$,

$$z = \frac{X - \mu}{\sigma} = \frac{725 - 750}{50} = -0.50$$



$$\begin{aligned} \text{Now, } P(675 < X < 725) &= P(-1.50 < Z < -0.50) \\ &= P(-1.50 < Z < 0) - P(-0.50 < Z < 0) \\ &= P(0 < Z < 1.50) - P(0 < Z < 0.5) \end{aligned}$$

By Symmetry

$$= 0.4392 - 0.1915$$

$$= 0.2419$$

The number of the person whose income is between RS 675 and RS 725

$$= N \times P$$

$$= 10,000 \times 0.2419$$

$$= 2419 \text{ persons.}$$

"Note to Remember"

- If the terms like: Fail, poor, Flat, Decrease, Below, under, Less than, Fewer than, Smaller than, Left hand Sides etc... are used in the question, then in that condition the given prob. should be plotted from " $-\infty$ " point.
- If the terms like: pass, Rich, Bump, increase, Exceed, over, Above, More than, Greater than, Right hand Sides etc... are used in the question, then in that condition the given prob. should be plotted from " $+\infty$ " point.

How to find z value from the given value of probability?

Ans: Follow the instruction given below:

$$\text{Let } p = 0.40$$

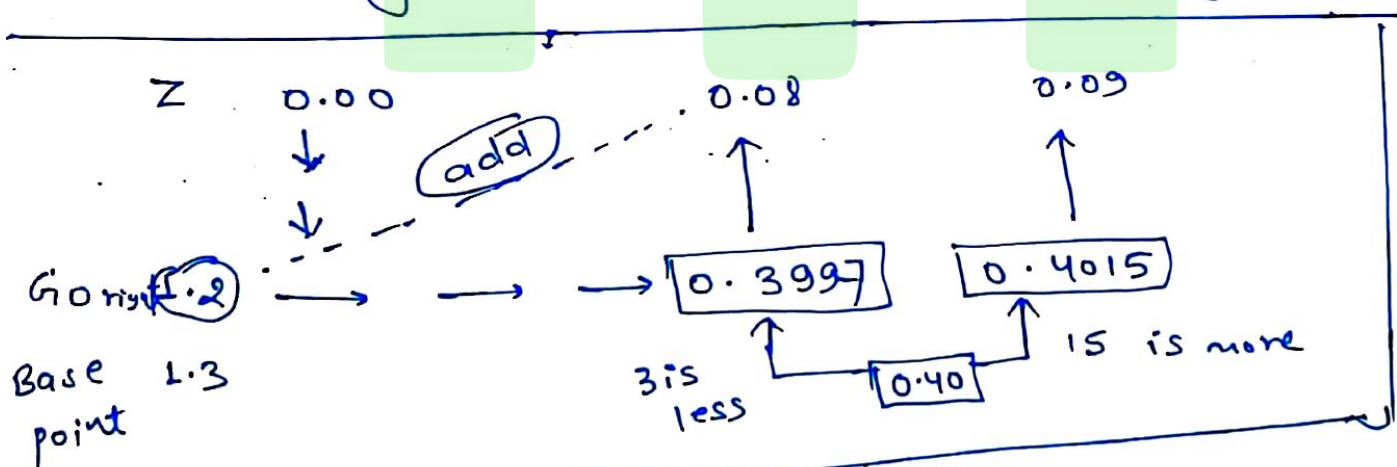
Step I: Go down vertically from 0.00 up to not meeting 0.40 value or more and mark the base point of z .



Step II: Select that z point which lies just above the base point and go right to find range for 0.40.

Step III: Select the nearest probability value with 0.40 and add the vertical point and horizontal point then get z value.

Searching z value from probability (Let $p=0.40$)



The prob. 0.40 approximately lies in the area of normal curve at $1.2 + 0.08 \Rightarrow 1.28$

$$\therefore z = 1.28$$

type-2

- Incomes of a group of 10,000 person were found to be normally distributed with mean RS 1520 and Standard deviation 160. find;
 - (a) Highest income of poorest 20% persons.
 - (b) Lowest income of richest 7000 persons.
 - (c) Limit income of middle 20% earners.

Solⁿ:

Total persons (N) = 10,000

Mean income (μ) = 1520

Standard deviation (σ) = RS 160.

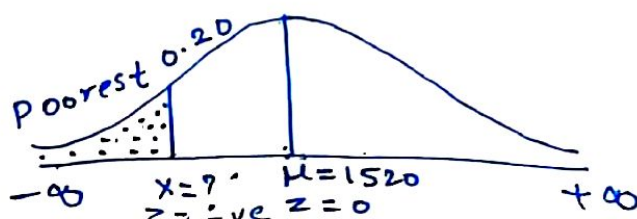
Note: જુન નામલે mean દિવાયકી છે તપ્પી નામમા પ્રશ્ન સૌધિયકી છે તને x સૌધિયકી હો તની વુઝનુ પદે.

Req. (a) Highest income of poorest 20%.

Highest income (x) = ?

$P(\text{poorest person}) = 0.20$.

in order to calculate the required value of x , the given prob. should be plotted into normal curve.



The prob. 0.30 approximately lies in the area of normal curve at $0.8 + 0.04 = 0.84$

$\therefore Z = -0.84$ (As it lies below mean)

Now,

$$Z = \frac{X - \mu}{\sigma}$$



$$\text{or, } -0.84 = \frac{X - 1520}{160}$$

$$\text{or, } -0.84 \times 160 = X - 1520$$

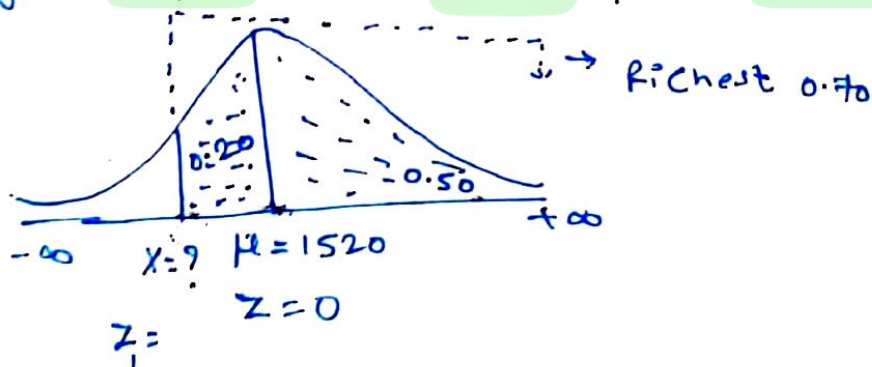
$$\text{or, } X = 1385.60$$

Req.(b) Lowest income of richest 7000

Lowest income (X) = ?

$$P(\text{richest person}) = \frac{7000}{10000} = 0.70$$

In order to calculate the required value of X, the given prob. should be plotted into normal curve.



The prob. 0.20 approximately lies in the area of normal curve at $0.5 + 0.02 = 0.52$

$\therefore Z = -0.52$ (As it lies below mean)

Now,

$$z = \frac{x - \mu}{\sigma}$$

$$\text{or, } -0.52 = \frac{x - 1520}{160}$$

$$\text{or, } -0.52 \times 160 = x - 1520$$

$$\text{or, } x = 1436.8$$

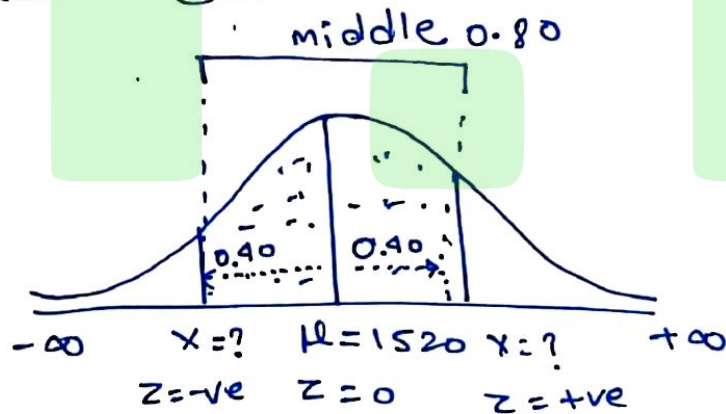
$$\therefore x = 1436.8$$

Req. (c) Limit income of middle 80%.

Limit income (x) = ?

$$p(\text{middle earners}) = 0.80$$

In order to calculate the required value of x the given prob. should be plotted into normal curve.



The prob. 0.40 approximately lies in the area of normal curve at $1.2 + 0.08$

$\therefore z = \pm 1.28$ (As it lies below & above mean) $= 1.28$
For lower limit ($z = -1.28$)

$$z = \frac{X - \mu}{\sigma}$$

$$\text{or, } -1.28 = \frac{X - 1520}{160}$$

$$\text{or, } -1.28 \times 160 = X - 1520$$

$$\text{or, } X = 1315.20$$



for upper limit ($z = 1.28$)

$$z = \frac{X - \mu}{\sigma}$$

$$\text{or, } 1.28 = \frac{X - 1520}{160}$$

$$\text{or, } 1.28 \times 160 = X - 1520$$

$$\therefore X = 1724.8$$

$$\begin{aligned} \text{Again, Range Income} &= 1724.80 - 1315.20 \\ &= 409.60 \end{aligned}$$

✓
✓
out of q.

→ frequency संगे statistics prob. है।

Q.3 In an examination, 15% of the students got less than 20% marks and 5% of the students got more than 75% marks. Assuming the distribution to be normal, find the mean & standard ~~normal~~ deviation.

Solⁿ

Here,

$$P(\text{Less than 20 marks}) = 0.15$$

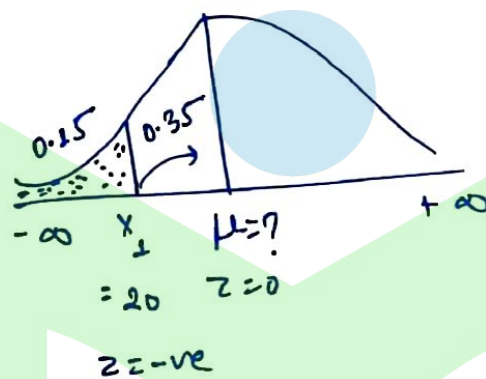
$$P(\text{more than 75 marks}) = 0.05$$

mean (μ) = ?

Standard deviation (σ) = ?

In order to calculate the required value of μ & σ , the given probability should be plotted into normal curve.

for less than 20 mark ($p=0.15$)



the prob. 0.35 approximately lies in the area of normal curve at $1.0 + 0.04$
 $= 1.04$

$\therefore Z = -1.04$ (As it lies below mean)

Now,

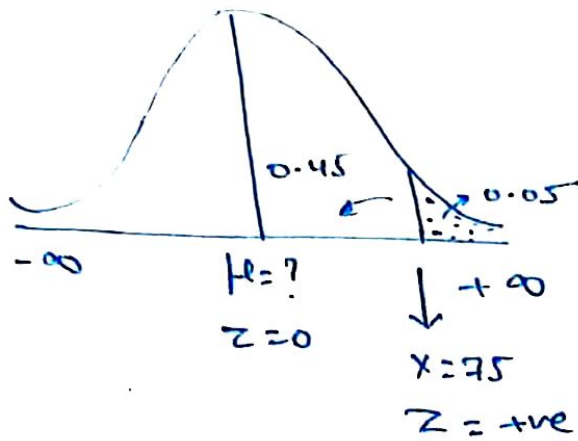
$$Z = \frac{X - \mu}{\sigma}$$

$$\text{or, } -1.04 = \frac{20 - \mu}{\sigma}$$

$$\text{or, } -1.04\sigma = 20 - \mu$$

$$\therefore \mu = 20 + 1.04\sigma \text{ ————— ①}$$

for more than 75 marks ($p=0.05$)



The prob. 0.45 approximately lies in the area of normal curve at $1.6 + 0.04$
 $= 1.64$

$\therefore z = 1.64$ (As it lies above mean)

Now,

$$z = \frac{X - \mu}{\sigma}$$

$$\text{or, } 1.64 = \frac{75 - \mu}{\sigma}$$

$$\text{or, } \mu = 75 - 1.64 \sigma$$

Solving eqⁿ ① & ②

$$\mu = \mu$$

$$20 + 1.04\sigma = 75 - 1.64\sigma$$

$$1.04\sigma + 1.64\sigma = 75 - 20$$

$$2.68\sigma = 55$$

$$\therefore \sigma = \frac{55}{2.68} = 20.52$$

from eqⁿ ①,

$$\mu = 20 + 1.04\sigma$$

$$\text{or, } \mu = 20 + 1.04 \times 20.52$$

$$\therefore \mu = 41.34$$

The required mean & S.D for the given data required.

T.U 2074

In a Statistic examination, 15% of the candidates scored more than 60 marks and 10% of the candidates scored less than 32 marks. Assuming that the marks were distributed normally find the mean marks and the standard deviation.

solⁿ

Here,

Let $X \sim N(\mu, \sigma^2)$

$$\text{Given } P(X < 32) = 10\% = 0.10$$

$$P(X > 64) = 15\% = 0.15$$

$$\text{then } \mu = ?$$

$$\sigma = ?$$

$$\text{SNV, } Z = \frac{X - \mu}{\sigma}$$

$$\text{Now, } P(X < 32) = 0.10$$

when, $x = 32$

$$z = \frac{32 - \mu}{\sigma} = -z_1 \text{ (say)}$$

$$p(z < -z_1) = 0.10 \quad \text{--- (i)}$$

$$\text{or, } 0.5 - p(-z_1 < z < 0) = 0.10$$

$$\text{or, } 0.5 - p(0 < z < z_1) = 0.10 \text{ [By Symmetry]}$$

$$\text{or, } p(0 < z < 0.1736) = 0.4$$

$$\therefore z_1 = 0.1736$$

from eqⁿ (i), we get

$$\frac{32 - \mu}{\sigma} = -0.1736$$

$$\mu = 32 + 0.1736\sigma \quad \text{--- (ii)}$$

$$\text{Again, } p(x > 60) = 0.15$$

when $x = 60$

$$z = \frac{60 - \mu}{\sigma} = z_2 \text{ (say)} \quad \text{--- (iii)}$$

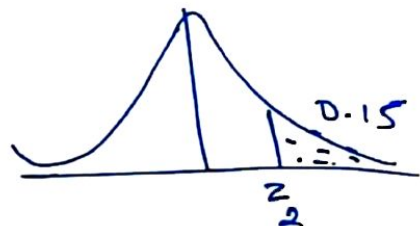
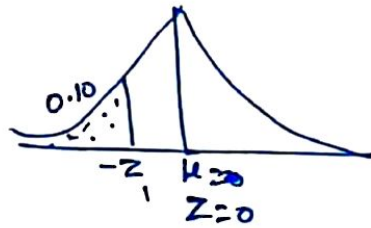
Now,

$$p(x > 60) = 0.15$$

$$\text{or, } p(z > z_2) = 0.15$$

$$\text{or, } 0.5 - p(0 < z < z_2) = 0.15$$

$$\text{or, } p(0 < z < z_2) = 0.35$$



$$P(0 < Z < 0.1368) = 0.35 \text{ (from normal table)}$$

$$\therefore Z_a = 0.1368$$

from (iii)

$$\frac{60 - \mu}{\sigma} = 0.1368$$

$$\text{or, } \mu = 60 - 0.1368\sigma \quad \text{--- (iv)}$$

Solving eqⁿ (iii) & (iv), we get

$$32 + 0.1736\sigma + 0.1368 = 60 - 32$$

$$\text{or, } 0.3104\sigma = 28$$

$$\sigma = 90.20 \approx 90$$

$$\therefore \sigma = 90$$

from eqⁿ (i),

$$\mu = 47.63 \approx 48$$

$$\therefore \text{mean}(\mu) = 48 \text{ \& \; S.D.}(\sigma) = 90$$

✓

• T.O 2075

State any three properties of normal distribution. The random variable x is distributed with mean $= \mu$ and variance $= 144$ and it is known that $P(X \geq 320) = 0.8438$. Find the value of μ .

Solⁿ

we given

$$\text{mean } (\mu) = 4$$

$$\text{variance } (\sigma^2) = 144$$

$$\therefore \sigma = 12$$

$$\therefore \text{standard deviation } (\sigma) = \sqrt{144} = 12$$

$$\therefore P(X \geq 320) = 0.8438$$

Now,

$$Z = \frac{X - \mu}{\sigma}$$

when, $x = 320$

$$\text{or, } Z = \frac{320 - \mu}{12}$$

$$\therefore P(X \geq 320) = P(Z \geq 320)$$

$$\text{or, } 0.8438 = \frac{320 - \mu}{12}$$

$$\text{or, } 0.8438 \times 12 = 320 - \mu$$

$$\therefore \mu = 309.87 \approx 310$$

$\therefore$ Hence the required value $(\mu) = 310$.

✓

Poisson Distribution



- It is a discrete probability distribution with infinite and unknown number of trial.
- It is propounded by French Mathematician Simon Denis Poisson in 1837.
- If the probability of occurring an event seems to be rare with at most 5% probability (i.e. $P \leq 0.05$) and more than 20 Sample Size (i.e. $n > 20$), then it is known to be Poisson Distribution.

Application of Poisson Distribution

Under following cases, we generally use Poisson distribution.

- Number of accidents taking place per unit of time on a road.
- Number of printing mistakes per page on a book.
- Number telephone coming per unit of time on a Switch board.
- Number of suicides reported per unit of time in a place.
- Number of twins' birth reported in a hospital.

- Let 'r' be the no. of occurring events, then the probability of getting 'r' events under poisson distribution is given by:

$$p(r) = \frac{e^{-\lambda} \lambda^r}{r!}, \quad r = 0, 1, 2, \dots, \infty$$

↓
No. of occurring events

λ = mean of poisson distribution (main parameter)

= Either

Given or

$n \cdot p$ or

$\frac{\sum Fx}{N}$

↓

↓

↓

with specific
time period
per hours, per
day etc.

$n > 20$ &
 $p < 5\%$

when
frequency
distribution
is given.

Ex When poisson?

1. If poisson is asked itself.
2. If events seems to be rare with at most 5% probability of occurrence having at least 20 sample.
3. If Mean or Average (i.e λ) is given with specific time period like per hour, per day, per 30 seconds etc.
4. If mean and variance values are equal.

Notes to Remember

- poisson distribution. મને પુરાં જાણતા આવતા λ જોડી હાલુ પદે । λ તાર્કિત કિત mean અવાલ Average. મને દિલ્લ કિત n & p તાર્કિ multiply ગરેર મિલ્લુ પદે ।
- poisson distribution મા સૌધિતે કુરાકો હામ & λ કો હામ સુતે હુનુ પદે ।



Some Expressions and Notations

Expression	Notation
None	$r=0$
Exactly 2	$r=2$
At least 2	$r \geq 2$
At most 2	$r \leq 2$
Not less than 2	$r \geq 2$
Not more than 2	$r \leq 2$
Less than 2	$r < 2$
more than	$r > 2$
within 2	$r \leq 2$
Exceeding 2	$r > 2$
Between 2 & 5	$2 < r < 5$

1) In a departmental store, the average number of customers arrived is 3 per hour. Find probability of (a) no customers; (b) Exactly 2 customers (c) at least 2 customers (d) at most 2 customers (e) customers between 2 and 6.

2) A telephone booth receives on average 4 calls in every 5 minutes. Find the probability of following question for the period of 5 minutes using suitable frequency distribution.

- (a) At least one call
- (b) More than 2 calls.
- (c) up to 3 calls.

(3) If 3% of an item produced by a factory is defective, find the prob. of exact 3 defectives in a sample of 50 items.

(4) The operations manager of a manufacturing company estimates that 2% of items are below the standard; find the prob. in a sample of 100 for (a) no item below the standard. (b) Two or more items below

item below the standard.

- 5] An observation of 2000 cars show that there is 0.02% prob. of having flat tires. By Using poisson distribution, determine the probability of not more than one car will have flat tires.



- 6] An automatic machine makes pins from the coil of iron wire. on average two in 400 pins are defective. If the pins are packed in a box of 100, what is the prob. that any given box will contain, (a) no defective (b) exactly three defectives, (c) one or more defective (d) less than two defectives (e) not less than 3 defective?

Solⁿ:

As per the information given in the question, it belongs to the poisson distribution.

we have,

$$p = \text{prob. of defective} = \frac{2}{400} = 0.005$$

$$n = \text{No. of pins in box} = 100$$

$$\text{Mean } (\lambda) = n \cdot p$$

$$= 100 \times 0.005 \Rightarrow 0.5$$

Let r be the no. of occurring defective pins, then the prob. of getting r under poisson distribution is given by

$$P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

(a) No defective ($r=0$)

$$P(0) = \frac{e^{-0.5} (0.5)^0}{0!} = 0.6065$$

(b) 3 defective ($r=3$)

$$P(3) = \frac{e^{-0.5} (0.5)^3}{3!} = 0.0126$$

(c) 1 or more defectives ($r \geq 1$)

$$r = 0, \overbrace{1, 2, \dots}^{\rightarrow} \infty$$

we have,

$$P(0) = 0.6065$$

now,

$$P_{\text{req. prob.}} = 1 - P(0)$$

$$= 1 - 0.6065$$

$$= 0.3935$$

(d) Less than 2 defective ($r < 2$)

$$r = \overleftarrow{0, 1}, 2, 1, \dots \infty$$

we have,

$$P(0) = 0.6065$$

$$P(1) = \frac{e^{-0.5} (0.5)^1}{1!} = 0.3026$$

now,



$$\text{req. prob.} = P_0 + P_1$$

$$= 0.6065 + 0.3026$$

$$= 0.9091$$

(e) Not less than 3 defectives ($X \geq 3$)

$$X = 0, 1, 2, 3, \dots, \infty$$

we have,

$$P(0) = 0.6065$$

$$P(1) = 0.3026$$

$$P(2) = \frac{e^{-0.5} (0.5)^2}{2!} = 0.0758$$

now,

$$\text{req. prob.} = 1 - (P_0 + P_1 + P_2)$$

$$= 1 - (0.6065 + 0.3026 + 0.0758)$$

$$= 1 - 0.9849$$

$$= 0.0151$$

✓

(2) A manufacturer of pins knows that on average 3% of its production is defective. He sells pins in boxes of 100 and guarantees that not more than 2 pins will be defective. What is the prob. that a box selected at random.

- (i) will meet the guaranteed quality?
(ii) will not meet the guaranteed quality?

Solution

Here,

Here, $p < 5\%$ and $n > 20$. So, it belongs to the Poisson distribution.

we have,

$$p = \text{prob. of defective} = 0.03$$

$$n = \text{No. of selected ~~workers~~ pins} = 100$$

$$\begin{aligned}\therefore \lambda &= \text{mean} = np \\ &= 100 \times 0.03 \\ &= 3\end{aligned}$$

Guarantee $\rightarrow$ not more than 2 pins will be defective.

$$P(x) = \frac{e^{-\lambda} \cdot \lambda^x}{x!}$$

$$\textcircled{1} p(\text{meeting guaranteed quality})$$

$$= p(\text{Not more than 2 defective})$$

$$= p(r \leq 2)$$

$$r = 0, 1, 2, 3, \dots \infty$$



we have,

$$p(0) = 0.0497$$

$$p(1) = 0.1494$$

$$p(2) = 0.2241$$

$$\begin{aligned} \therefore p(r \leq 2) &= p(0) + p(1) + p(2) \\ &= 0.0497 + 0.1494 + 0.2241 \\ &= 0.4231 \end{aligned}$$

$$\textcircled{11} p(\text{not meet the guaranteed quality})$$

we know that

$$\begin{aligned} &p(\text{not meet the guaranteed quality}) \\ &= 1 - p(\text{meet the guaranteed quality}) \end{aligned}$$

$$= 1 - 0.4231$$

$$= 0.5769$$

Q. A manufacturer finds that the average demand per day for the mechanics to repair the product is 2.5. The no. of mechanics employed is 2. In how many days in three months (a) Both mechanics would be free; (b) Some demand is refused.

Soln:

As per the information of average demand per day given in the question, we need to apply poisson distribution for required probability.

we have,

$$\lambda = \text{Average demand} = 2.5 \text{ per day}$$

$$\text{No. of mechanics employed} = 2.$$

$$\therefore P(r) = \frac{e^{-\lambda} \cdot \lambda^r}{r!}$$

Now,

(a) Both mechanics would be free

(যদিও demand zero হলে তখনই)
mechanics free হলে

If there is no any demand (i.e. $r=0$), then both mechanics would be free. So the required prob. can be expressed as $P(0)$:

$$P(0) = \frac{e^{-2.5} (2.5)^0}{0!} = 0.0821$$

So, the required no. of days in three months on which both mechanics would be free.



$$= N \times 0.0821$$

$$= 90 \times 0.0821$$

$$= 7 \text{ days.}$$

⑥ Some demand is refused.
(demand स्वीकार नगर्ना)

There are two mechanics and the question is silent about the capacity of mechanics. It means two mechanics can cover two demands at a time (1:1 ratio)

So if there are more than 2 demands than those demands should be refused and hence the required prob. can be expressed as

$$P(r > 2)$$

$$r = 0, 1, 2, \overset{P=?}{\overbrace{3, \dots \infty}}$$

$$P(r > 2) = 1 - [P(0) + P(1) + P(2)]$$

$$= 1 - 0.5439$$

$$= 0.4561 \checkmark (\text{refused \text{सिवा} prob. \text{है}})$$

< So, the required no. of days in three months on which some demand is refused

$$= N \times p(1 > 2)$$

$$= 90 \times 0.4564$$

$$= 41 \text{ days.}$$

(Imp) Qn

Vehicles arrive an intersection at the rate of 10 per minute. A traffic light cycle lasts 45 seconds. If the number of vehicles that arrive per cycle follows a poisson distribution what is the probability that exactly 6 vehicles will arrive in a cycle.

यदि cycle में लगी कारों की दर $\xrightarrow{\text{per min}} \xrightarrow{\text{per seconds}}$

(per min 10 veh आसकत)

Solⁿ

Here,

As per the requirement of the question we need to apply poisson distribution for required probability.

Here, the prob. is asked for per cycle and traffic light cycle ends in 45 seconds. Thus the vehicles arrival rate per minute should be converted into per cycle (i.e. per 45 seconds).

60 seconds (i.e. 1 min) $\rightarrow$ 10 vehicles arrived

$$1 \quad \dots \quad \rightarrow \quad \frac{10}{10^n} \quad , , \quad , ,$$

45 seconds (one cycle) $\rightarrow \frac{10}{60} \times 45$ vehicles arrived

= 7.5 vehicles arrived

$\therefore \lambda = \text{Average arrival rate} = 7.5 \text{ per cycle.}$

$$p_0' = \frac{\epsilon \cdot \lambda^x}{\delta_0'}$$



→ (exactly 6 vehicles per cycle)

$$P(6) = \frac{e^{-7.5} \cdot 7.5^6}{6!}$$

$$= 0.1367$$



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